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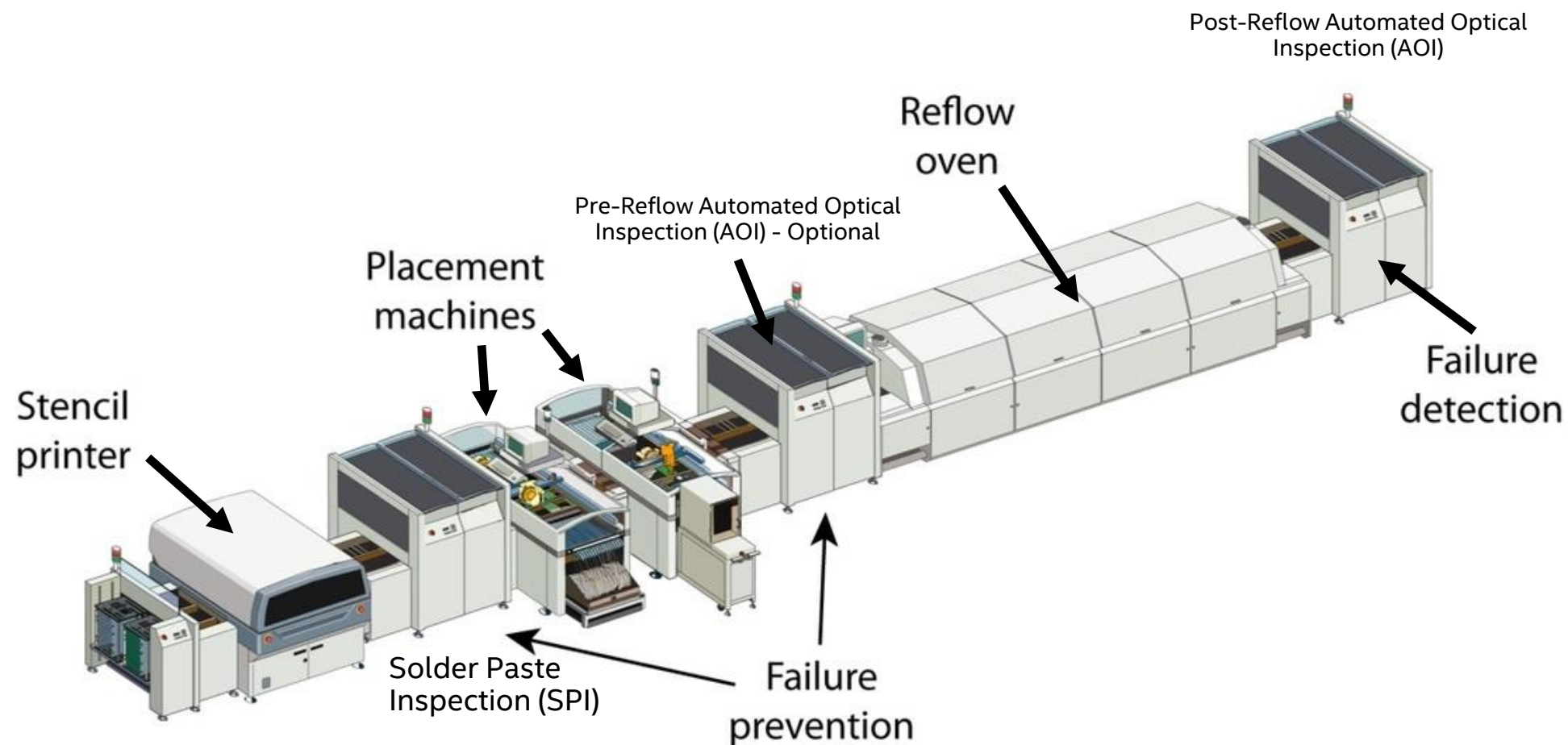
2021 MTL Customers CRD Summary Report

Rev1.0, CQ&R CMEE Team, 2021Q3



intel®

General SMT Line Layout



Objectives

Share Findings From CMEE Mobile Industry Data Collection (MTL Platform Customer Survey)

- Data Collected From 10 ODM Customers/Sites
 - Compal (KunSan – Build for Dell/HP, ChengDu – Build for Dell, ChongQing – Build for Acer)
 - HuaQin (Build for HuaWei/Asus/Lenovo/Samsung)
 - Inventec ChongQing (Build for HP/Asus)
 - LCFC (Build for Lenovo)
 - MSI KunSan (Own Brand)
 - Pegatron (ChongQing – Build for Asus)
 - Quanta (ChongQing – Build for HP)
 - Wistron TaiWan (Build for Lenovo)

Highlights

- Due to most customers aren't in LTS HVM yet, thus for a lot of LTS related questions customer answered using SAC experience
- T4 still the major paste customer is using. And step stencil is widely accepted in the industry ➔
- Only 3/10 customers have thermal cure adhesive process. Anticipate high customer impact if must use thermal cure process ➔
- Thunderbolt and SD WIFI module are the most common Intel branded component used by customer ➔
- Customers are worrying about MTL M max temp 200C because they afraid during rework process the component body temp will over the limit ➔

Solder Paste Printing

Solder Paste Printing (1 Of 6)

Have you had any prior experience printing paste 0.37mm pitch BGA?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
No	No, min 0.4mm	No	Yes	Yes 0.35mm pitch CSP
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Lenovo: No ThinkPad: Yes, 0.35mm	No	Yes	No, min 0.4mm	Yes, 0.35mm
Which Stencil tension type & values are you using for 0.37mm pitch BGA?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
50N	32-35N	>35N	45N~50N	>=30N
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
N/A	Not using 0.37mm BGA now	38~42N	30~50N	32~38N

Half of the customer have small pitch (<0.4mm) print experience

Solder Paste Printing (2 Of 6)

* Red means the paste is not in our recommendation list

List top 3 LTS pastes and percentage of each type planned to use in your production line for MTL-M				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
NA	NA	Nihon B37	Eunow DW1000 (still testing)	NA, current testing Eunow, ShenMao and Koki
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Lenovo: 金讯EcoLM-C15-GL, Bi 49%-53%. (100%) ThinkPad: Senju L28-N143HF (100%)	NA	Current testing: 1.Eunow DW1000 2.YunNanTin YT-AB02 (57% Bi)	Shenmao PF735-PQ10; Aojiang HN-302; Compo LF143S-PF.	Shenmao: PF735-PQ10-10 Sn-48Bi-Ag-X (Intel) Senju: L29-145HF Sn-58Bi-Sb-Ni (Evaluating) Senju: L28 -N143HF Sn-35Bi-0.5Cu-0.03Ni (Lenovo)

Few customers using LTS paste outside Intel recommendation list. Most still near eutectic, few of them are none-eutectic

Solder Paste Printing (3 Of 6)



For MTL-M BGA package with 0.37 mm pitch, what is solder paste mesh size planned to be used?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
T4	T4	T4	T5	T4.5 or T5
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
T5	NA	T4	T4	T4
For 0.37mm BGA pitch, will you use a single stencil thickness or a stepped stencil?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Use Step stencil	Most situation use step stencil (for connectors)	Few situation use step stencil	Few situation use stepped stencil	Most situation use step stencil
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Use step stencil	Yes, 100% use step stencil	Most situation use step stencil	Few situation use stepped stencil	Not use step stencil

**T4 is still the major paste type;
More and more customer now using stepped stencil then before**

Solder Paste Printing (4 Of 6)

If a single stencil thickness (mm or mils) is used for 0.37mm BGA pitch, what is your thickness preference?

Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
3mil	3mil	3mil	3 mil	3mil
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
0.06mm (~2.5mil)	NA	3mil	0.06mm (~2.5mil)	3mil

When using step down stencils, what is max print (squeegee) speed (mm/sec)?

Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
200mm/s	100mm/s	80mm/s	80 mm/s	120mm/s
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
150mm/s	70 ±20mm/s	80 ~ 120mm/s	120mm/s	Same as no step

When using step down stencils, what is print (squeegee) pressure (kg)?

Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
8kg	0.5kg per 25mm squeegee length	7kg	No answer	11Kg
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
0.26Mpa(Hitachi)	12±4kg	8 ~ 12kg	10Kg	Same as no step

3mil stencil still most commonly used for <= 0.4mm BGA

Solder Paste Printing (5 Of 6)

For 0.37mm pitch BGA, what is the smallest AR (Area Ratio) capable in HVM				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
0.69	0.7	0.62~0.7	0.625	0.66
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
0.7	NA	0.7	> 0.7	0.68
Is nanocoat technology used on stencil for 0.37 mm pitch BGA				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Will consider it	Not using now	Will consider it	Yes	Yes
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Yes	NA	Fine Grain always, and some projects use nanocoat	Yes	Yes
How many strokes for kneading (per print direction) when a new paste is applied onto the stencil? Please also state if different for LTS vs. SAC.				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
2	4	3~5	6	2
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
No	2	2	No	1~3

Solder Paste Printing (6 Of 6)

What is the auto cleaning frequency / setting of the stencil (how many print strokes before stencil is cleaned)? Please also state if different for LTS vs. SAC.				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
8	6	6~10	4	6
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
15	4	3 ~ 6	6	Min: 1~3
Please describe under what conditions boards are either washed / cleaned manually (IPA) versus Ultrasonically?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Manual	Manual	Manual	Manual	Manual (IPA)
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Manual IPA	Manual first, then use equipment	Manual	Ultrasonic	Special equipment

Most customers still use manual clean process for mis-printed boards

Solder Paste Inspection (SPI)

Solder Paste Inspection (1 of 2)

What are your SPI volume control limits (+/- %) for 0.37 mm pitch BGA				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
50% - 150%	70% - 165%	80% - 140% (LTS)	40% - 160%	60% - 160%
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
50% - 140%	NA	50% - 180%	50% - 150%	Follow customers
For SPI volume control limits with 0.37mm pitch BGA, what is percentage of defects?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
NA	NA	NA	NA	NA
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
NA	NA	NA	NA	NA

Customers still follow their standard SPI control limits even for small pitch BGA similar to prior survey

Solder Paste Inspection (2 of 2)

What are your SPI X (um) and Y (um) offset limits for 0.37 mm pitch BGA?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
20%	50%	30um	33 um	70um
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
50um	NA	40%	100um	Follow customers
What is the SPI tool threshold setting values (um) used in the algorithm?				
[FYI: The threshold value is the minimum height above which the volume is measured. The threshold value is measured from either the Cu or from the Soldermask (tool decides based on the pitch)].				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
35um	NA	NA	4um	30um
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
50um	Confidential	NA	NA	Follow customer's request

Pick and Placement (PnP) Machine

Pick and Placement (PnP) Machine (1 Of 1)

State BGA Placement Force in Newtons when Intel MTL-M 0.37mm pitch BGA is placed?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
2.5N	2.5N	2.5N	2.5N	2.5N
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
2.5N	NA	NA, Fuji NXT	2.5N	2.5N
Under what conditions do you decide to use a whole process pallet?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Current no such requirement	1. No breakout process edge 2. PCB T<0.8mm	Current no such requirement	PCB T<0.8mm	PCB T<0.6mm
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
1. No breakout process edge 2. PCB T<0.6mm	Current no such requirement	PCB T<0.8mm	PCB T<0.7mm	Depends on product size, weight and PCB thickness

Customers use whole process pallet in 2 conditions:

1. There is no breakout process edge designed for the PCB
2. PCB T < 0.6~0.8mm

Reflow Convection Oven

Reflow Convection Oven (1 of 3)

What will be your Peak Temperature Range for LTS BGA?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
165~175C	NA	185~200C	165 - 180C	170C - 185C
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Lenovo:180+/-5C ThinkPad:190-195C	NA	160-180C	Shenmao, 155-185C; Aojiang HN-302, 160-180C; Compo LF143S-PF, 180-200C.	Depends on paste PF735-PQ10-10 165 - 175C L28 -N143HF <190C

The highest peak reflow of LTS process <=200C

Reflow Convection Oven (2 of 3)

Do you measure the cooling slope for reflow profile for SAC? If yes, what is the criteria for passing?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
-4~-1C/S	-3~-1C/S	-3~-1C/S	-3~-1C/S	-3~-1C/S
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
-3~-1C/S	-3~-1C/S	-3~-1C/S	NA	Depends on product
Do you measure the cooling slope for reflow profile for LTS? If yes, what is the criteria for passing?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
NA	NA	-3~-1C/S	-3~-1C/S	-3~-1C/S
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Lenovo: -4~-1C/S ThinkPad -3~-1C/S	NA	-5~-1C/S	Shenmao, -6~-1C/s; Aojiang HN-302, -8~-3C/s; Compo LF143S-PF, ≤-4C/s.	-3~-1C/S, <115C out of reflow oven
- If yes, how do you calculate the cooling slope for LTS?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
NA	NA	NA	Max slope over 20 seconds down to 120C	Max slope over 60 seconds down to 110C
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Max slope over 65 seconds down to 100C	NA	NA	NA	NA

Reflow Convection Oven (3 of 3)

What changes will be made to reflow oven to accommodate an LTS reflow profile?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
NA	NA	Change heating zone temp setting and belt speed	Change heating zone temp setting and belt speed	Change heating zone temp setting and belt speed
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Change heating zone temp setting	NA	Change heating zone temp setting and belt speed	NA	Change heating zone temp setting
Do you plan to follow all of the Intel MAS for LTS reflow parameters?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Yes	NA	Yes	Yes	Yes
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
No, we will follow paste vendor suggestion	NA	Yes	Yes	Yes, will consider Intel MAS and paste vendor suggestion
What is the Belt Speed (cm/min) used for an LTS profile?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
NA	NA	90cm/min	120 cm/min	125 cm/min
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
140cm/min	NA	100cm/min	NA	Depends on product

Adhesive SJR (Solder Joint Reliability)

Adhesive SJR (1 of 5)

How do you decide whether to use board level adhesives on LTS?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Always use UV corner glue for big BGAs	Always use UV corner glue for big BGAs	Always use UV corner glue for big BGAs (CPU, GPU, SSD)	Use corner glue all the time	Follow customer requirement + drop test result
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Always use underfill for > 10x10mm BGAs (CPU/GPU/RAM/PCH)	NA	NA	Always use corner glue for key components. Current evaluating SJEM process	CPU+Customer requirement+test fail location+high stress area
Do you follow Intel's MAS dispense & KOZ recommendations for Client Mobile Products with regards to adhesive material?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Yes	Yes	Yes	Yes	Yes
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Yes	Yes	Yes	Yes	Depends on customer requirement
What adhesive brand and model do you use for LTS?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
NA	NA	UV glue	LOCTITE 3705 (UV)	NA
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
ASEC 3917 Syscotech(新懿) AS3109A7 (Corner Underfill)	NA	SeaYu(信友) UV3329 (UV)	NA	Loctite 1172A (Intel required underfill)

Adhesive SJR (2 of 5)

Do you apply adhesives to other Intel branded components (Eg: PMIC, Thunderbolt, Wifi, Clover Falls, etc...) on motherboards manufactured with LTS pastes? (Answer "NA" if not using other Intel branded components)				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
NA	NA	Depends on dispense spacing	No	NA
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Yes, PCH	Yes, PCH	Yes, PCH	Yes, PCH	No
Do you apply adhesives to other non-Intel BOM components (Eg: Memory CSP, Connectors, etc...) on motherboards manufactured with LTS pastes?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
NA	NA	Yes, Memory	No	NA
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Yes, GPU/Memory	No	Yes, GPU	Yes	Yes, GPU, memory connector, WLAN connector)

CPU, PCH, GPU, Memory are the most common components that customers apply adhesive onto

Adhesive SJR (3 of 5)



Do you have HVM thermal cure glue process capability in your factory?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
No	No	No	Yes	Yes
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
No	No	No	No	Yes
- If not, what would be the impact to setup thermal cure glue process in future?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Equipment investment, Adhesive Qualification, No enough factory floor space	NA	Equipment investment, Adhesive Qualification, Line reconfig	NA	NA
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Equipment investment, Line reconfig	Unlikely	NA	Line reconfig, Unlikely to implement because heat cure will cause PCBA deformation	NA

Only 3/10 customers have thermal cure adhesive process in their factories. Unlikely implement thermal cure process w/o big impact

Adhesive SJR (4 of 5)

Do you have HVM underfill/corner fill process capability in your factory?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
No	No	No	Yes	No
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Yes	No	No	No	Yes
- If not, what would be the impact to setup underfill/corner fill process in future?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Equipment investment, Adhesive Qualification, No enough factory floor space	NA	Equipment investment, Adhesive Qualification, Line reconfig	NA	Equipment investment, Line reconfig
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
NA	Unlikely	NA	Line reconfig, Unlikely to implement because heat cure will cause PCBA deformation	NA

Only 3/10 customers have underfill/corner fill process in their factories, because they build for Lenovo

Adhesive SJR (5 of 5)

Do you apply adhesive on BGAs right after SMT/before any test? If yes, what is the reason?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Yes, easy to apply glue before depanel	Yes, to prevent SJ crack	NA	No	Yes, to prevent SJ crack
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
No	Yes, easy to apply glue before depanel	Yes	Yes	No
Do you have Pre-Reflow In-Line adhesive dispense capabilities, to hold select SMT parts from falling off the board on the 2nd Pass through SMT?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Yes	Yes	Yes	No	Yes
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Yes	Yes	No	No	Yes

- 2 reasons customer apply adhesives right after reflow:**
- 1. Easy to process before the board was de-panel-ed**
 - 2. Prevent solder crack on the following mfg process**

X-Ray FA Capability

X-Ray FA Capability (1 of 1)

State Xray (Failure Analysis) Supplier Name, Model# and Year of Manufacture, used for Intel BGA?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Nordson-XD7500VR	Dage -XD7500/2007Y	NA	phoenix xlaminer MXCN000619-332618 March 2019	SEC X-eye SF160 2013/04/25
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Nordson/Dage-Quadra6-Year2018	Old machine, 15 years ago	YXLON Y.Cougar.SMT 2014/12	Nordson/Dage-Quadra7	Dage & TRI

Rework

Rework (1 of 5)

What is the Rework Tool used for LTS BGA formulation?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
QUICK EA-H15	QUICK EA-H15	QUICK EA-A10	Ersa HR 600-2	业欧 PFT-950/业欧 TP-500
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Ersa HR 600-2	NA	Hipo Tech RD500-III; HBR-RD500S	No answer	SRT2200i
Does the Rework tool use Convection Air or Infrared Air for heating package/board during reflow?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
IR	IR	Both	Convection Air	Top: Convection Air + Bottom IR
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
IR	NA	Convection air	IR	Convection air
What tool is used for site dress of pad site after component removal for 0.37 mm pitch Intel BGA package?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
4C houseshoe iron tip	4C houseshoe iron tip	Iron tip	4C houseshoe iron tip	Auto scavenger tool
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
4C houseshoe iron tip	NA	4C houseshoe iron tip	No answer	Iron tip

Rework (2 Of 5)

What is the iron tip temperature setting for site dressing of LTS BGA rework?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
420C (SAC)	370-390C (SAC)	380C (SAC)	250C	250C
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
390±5C	NA	250C	NA	350C
When is paste or Flux used for Intel SOC for LTS BGA?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Print paste to BGA ball + apply flux to PCB pad	Paste	<0.4mm, use flux, >= 0.4mm, use paste	Paste	Paste
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Paste	NA	<0.35mm, use flux, >= 0.35mm, use paste	<0.4mm, use flux, >= 0.4mm, use paste	Paste

Most customers still capable to print paste for BGA pitch = 0.4mm

Rework (3 Of 5)

Please state LTS Paste Vendor Name and Formulation name for BGA rework?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
YunNan Tin YW9-3005-98CY04 (SAC)	YunNan Tin YW9-3005-98CY04 (SAC)	Nihon B37	N/A (Current is testing Eunow and Shenmao)	N/A (Current testing Eunow, Shenmao and Koki)
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Lenovo: 金讯EcoLM-C15(100%) ThinkPad: Senju L28-N143 (100%)	NA	NA	NA	Same as SMT paste
Please state LTS Flux Vendor Name and Formulation name for BGA rework?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
AIMSOLDERPASTE NC254 (SAC)	AIMSOLDERPASTE NC254 (SAC)	NA	NA	NA
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
AMTECH-559	NA	Eunow PASTE FLUX 100G/B07 (SAC?)	NA	Same as SMT paste

Rework (4 of 5)

Do you plan to follow all of the Intel MAS for LTS rework parameters?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Yes	Yes	Yes	Yes	Yes
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Yes	NA	Yes	Yes	Will follow paste vendor recommendation and Intel MAS
Cycle Time (average) in mins to replace/remove LTS BGA?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
30min	20-25min	25min	20min	30min
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
25min	NA	23min	25min	30min
Have you had success reworking with solder pastes at your facility for BGAs >0.35mm and <=0.4mm pitch or had any prior experience working with BGAs >0.35mm and <=0.4mm pitch?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Yes	Yes	No	Yes	Yes
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Yes	Yes	Yes	Yes, success rate 70%~80%. 0.4mm (~800 pins)	Yes

Rework(5 Of 5)

If N2 is needed for rework, what is the factory impact of installing it?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Not likely	Need new equipment	Not likely	Need new equipment	Need new equipment, not likely now
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
NA	NA	NA	NA	NA
Which flux type do you prefer for reworking full stack LTS BGAs with >0.35mm and <=0.4mm pitch?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
RF	RF	Use paste instead of flux	RMA	NA
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
RF	NA	RMA	NA	RF

Only very few customers are capable for N2 rework (Pls refer to ADL CRD)

Large EMI Shields (> 40 x 40 mm)

Large EMI Shields(1 of 1)

For an LTS process, do you utilize any solder additive process for the EMI shield (>40 x 40mm size) in reflow to increase mechanical stability?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Step stencil	NA	Step stencil, enlarged stencil apertures, top hat on reflow pallet	Step stencil	Step stencil, small solder chip additive
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
NA	Step stencil	Step stencil and/or enlarged stencil apertures	NA	NA

Step stencil is the most common method for solder paste additive

Bare PCB (Printed Circuit Board)

Bare PCB (1 Of 1)

Is your entire motherboard to be built with LTS paste?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Will consider it	NA	Yes	Yes	Yes
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Yes	NA	Yes	Bottom SAC, Top LTS	Yes
Are there plans, or a desire, to surface mounting the MTL-M (having LTS ball) with SAC pastes (reverse hybrid)?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
NA	NA	No	No	No
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
No	NA	No	NA	No currently, If Intel has the data pls share

LTS (Low Temperature Solder)

Low Temperature Solder (1 of 3)



What are your LTS preferred paste formulation?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
NA	NA	Hypoeutectic Bi% (35-40%)	Near Eutectic Bi% (58%)	Near Eutectic Bi% (58%)
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Near Eutectic Bi% (58%)	NA	Depends, haven't decided yet	Depends, haven't decided yet	Near Eutectic Bi% (58%)
Are you planning to use Intel Branded Components other than MTL-M SOC on your motherboards?(If Yes, please answer the next question)				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Yes	No	Yes	Yes	Yes
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Yes	Yes	Yes	Yes	NA
- What other Intel branded components are you planning to use on your motherboards?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
WIFI	NA	Thunderbolt, WIFI	WIFI	WIFI
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
PMIC, Thunderbolt, WIFI	WIFI	WIFI	Thunderbolt, WIFI	N/A

Low Temperature Solder (2 of 3)



What is the maximum board thickness you will surface mount MTL-M BGA onto?

Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
1.2mm	NA	1.0mm	1.0mm	1.2mm
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
1.2mm	1.2mm	1.0mm	0.7~0.8mm	1.2mm

Are there any other component suppliers providing full stack LTS BGAs?

Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
No	NA	No	No	No
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
No	No	No	No	No

Do you have any concerns if the maximum peak reflow temp for LTS, supported in the MTL-M MAS, is 200C?

Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Yes	NA	Yes	No	Yes
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Yes	No	No	No	No

ODM is worrying about the package body temp will over 200 C during rework process. Wistron measurement shows BGA body temp at 202c (peak temp 200C) during LTS rework. Consider some LTS paste will require peak temp up to 220C, Wistron's preference of MTL M max temp is 230C.

Low Temperature Solder (3 of 3)

Based on your available data, what is the risk level to implement LTS process on MTL-M product(s)? :

Please select one:

A. Low and ready.

B. Medium and still need to collect some data

C. High and limited data available

D. Show stopper

Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
D. Show stopper	D. Show stopper	B. Medium and still need to collect some data	C. High and limited data available	B. Medium and still need to collect some data
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
A. Low and ready.	C. High and limited data available	B. Medium and still need to collect some data	D. Show stopper	A. Low and ready.

If answer is B, C or D, please provide what data is required to bring risk level down to Low?

Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
SJO, SJ crack	NA	SJ strength compared to SAC	Dye Pry Humidity Test Corner Drop Test Thermal Shock Test Fragility Shock and Vibration Torture Test I/O connector strength and durability Weighted Vibration Test Weight Drop Test	SMT setting and SJ reliability test
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
During rework the CPU surface temp will > 200C	Component max temp, HT warpage, SJ reliability	Still under test	No answer	NA

SD WLAN

SD WLAN (1 of 4)

What is the percentage of your design having the SDWLAN on the opposite side with major BGA (CPU, Gfx etc)? Please provide a rough % number.				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
30~40%	NA, not using SDWLAN	10%	10%	0%
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
0%	10%	0%	20%	Not using SDWLAN
Where is the SDWLAN located in your board design? Please answer: Center, Edge, Corner (Refer to the picture-->)				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Edge	NA	Edge, corner	Corner	Edge, Corner
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Edge	Corner	Edge, corner	Edge, corner	
In your design, what is the closest distance from board edge/corner to SDWLAN module?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
1mm	NA	>5mm	1.5mm	1mm
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
5mm	5mm	0.5mm	20mm	

SD WLAN (2 Of 4)

What is the % of your projects are using 3mil stencil and what the % of 4mil stencil for SDWLAN solder down?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
100% 4mil	NA	100% 4mil (~100um)	100% 4mil (~100um)	20% 5mil (~127um), 80% 4mil (~100um)
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Lenovo:80% 4mil, 20% 5mil ThinkPad:0.11mm 80% 0.1mm 20%	90% 4mil	100% 4mil	100% 4mil	
Do you need to have a 3mil stencil thickness recommendation in the Intel MAS?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Yes	NA	No	Yes	Yes
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Yes	Yes	Yes, we use more and more 3mil stencil, step stencil will impact print quality, we prefer to have 3mil recommendation	No	

Current 4mil and 5mil stencil still the mainstream for SDWLAN, but big portion of customers are hoping to have 3mil stencil recommendations

SDWLAN (3 of 4)

Do you scan the SN# of Intel SDWLAN module, and link it with board SN# and save it into your database?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
No	NA	No	No	No
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
No	No	No	No	
- If yes, can you provide the specific scanner (brand, model) you are using and what is the distance/angle when you scan?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
No	NA	No	No	No
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
NA	NA	NA	NA	
Are you aware of the Intel SDWLAN new land pattern recommendation? To date, how many of your designs had implemented this new land pattern?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Yes, haven't implement yet	NA	Will implement for all future project	Yes, 70% current design and 100% all future new design	Yes
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Yes, 7% existing projects, 100% for future new projects	Unclear	Yes, 3 projects now. 100% for future projects	Yes	

SDWLAN (4 Of 4)

Do you have any suggestions to Intel to further improve the PCB land pattern design?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
No	NA	No	No	No
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
No	SDWLAN warpage after reflow causing SJO and SJ short, it is keeping on bothering us	No	Less via hole on the footprint	
Do you have any suggestions to Intel to further improve the SDWLAN module design aimed to achieve higher SMT yield?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
No	NA	No	No	No
LCFC	MSI	Pega CQ	Quanta CQ	Wistron
Recommend LCC type package	No	No	IO extend to substrate edge	

